

Answer

Test 2 Chemistry II

Semester 3 2018/2019

Question 1 (10 Marks)

1. The conjugate base of H_2AsO_4^- is HAsO_4^{2-} (D)



Acid

Conjugate
Base

Bronsted-Lowrey acid or base Theory
Acid has an extra H^+ in its formula

Question 1 (10 Marks)

2. In the reaction, $\text{HClO}_3 + \text{N}_2\text{H}_4 \rightleftharpoons \text{ClO}_3^- + \text{N}_2\text{H}_5^+$, which one of the sets below list both of the base species involved in the equilibrium? **(D)**

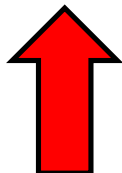


Acid

Base

Conjugate
Base

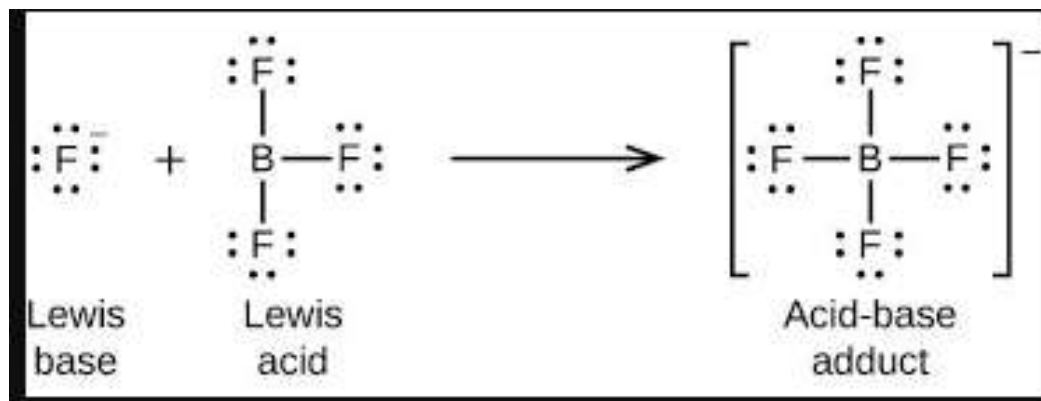
Conjugate
Acid



Question 1 (10 Marks)

3. F^- ion can directly react with BF_3 to form BF_4^- anion. In this reaction, **(C)**.

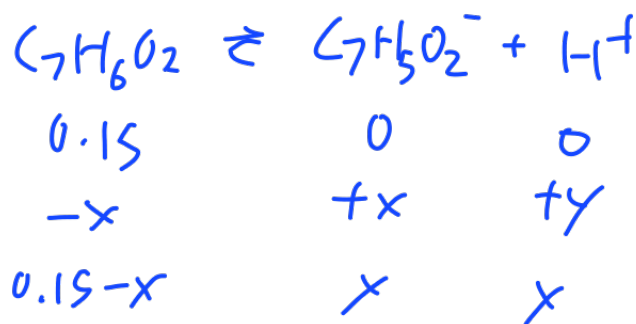
BF_3 molecule acts as Lewis Acid, accepting an electron pair from F^- ion to form a coordinate covalent bonds.



Question 1 (10 Marks)

4. The K_a for ^{weak acid} benzoic acid, $C_7H_6O_2$ is 6.28×10^{-5} . Calculate the pH of 0.15M solution of the acid.

(B) 2.52

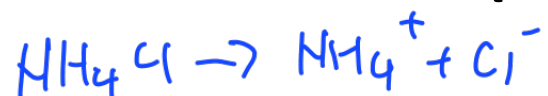


$$K_a = \frac{x^2}{0.15-x} = \frac{x^2}{0.15} = 6.28 \times 10^{-5}$$

$$x = 3.07 \times 10^{-3} = [H^+]$$

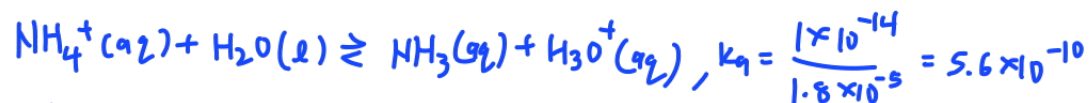
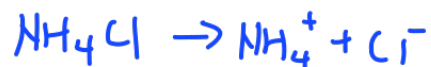
$$\begin{aligned} pH &= -\log [H^+] \\ &= -\log (3.07 \times 10^{-3}) \\ &= 2.51 \end{aligned}$$

Question 1 (10 Marks)



5. Calculate the pH of a 1.0M solution of $\text{NH}_4\text{Cl}(\text{aq})$. The K_b for NH_3 is 1.8×10^{-5}

Answer (B) 4.63



$$5.6 \times 10^{-10} = \frac{x^2}{1-x} = x^2$$

$$x = [\text{H}^+] = 2.366 \times 10^{-5}$$

$$\text{pH} = -\log [\text{H}^+] = 4.63$$

Question 1 (10 Marks)

6. A Galvanic cell has two electrodes. Which statement below is correct? **(B)**

Similarities and differences between the galvanic (voltaic) cell and the electrolytic cell

	Voltaic/Galvanic cell	Electrolytic Cell
Spontaneity:	Spontaneous	Non-spontaneous
E_{Cell}° :	Positive (+ve)	Negative (-ve)
Cathode:	(+ve) electrode Reduction process	(-ve) electrode Reduction process
Anode:	(-ve) electrode Oxidation process	(+ve) electrode Oxidation process
Flow of e:	Anode to cathode	Anode to cathode
Flow of ions:	Anions to anode Cations to cathode	Anions to anode Cations to cathode

Question 1 (10 Marks)

7. An electrochemical cell consists of $\text{Cu(s)}|\text{Cu}^{2+}(\text{aq})$ half cell and $\text{Zn(s)}|\text{Zn}^{2+}(\text{aq})$ half cell connected by a salt bridge. Oxidation occurs in the zinc half cell. The cell can be represented in cell notation as: **(D)**

A 'cell diagram' or 'shorthand notation' describes a voltaic cell with the use of symbols

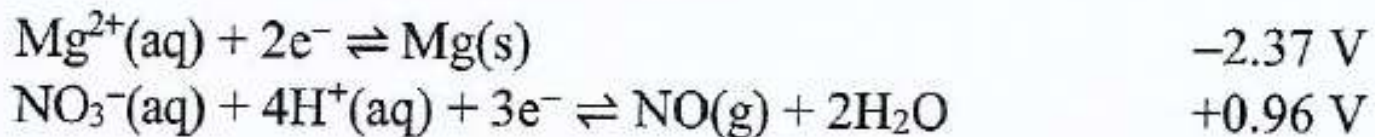
The 'shorthand notation' describing the Zn-Cu cell above is



The ' | ' symbol separates the electrodes from the electrolytes and ' || ', symbolic of the salt bridge, separates the half-cells

Question 1 (10 Marks)

8. Using SRP Given Below



calculate the value of E°_{cell} with the reaction,

Answer: **C**

Question 1 (10 Marks)

9. When an aqueous solution of sodium chloride is electrolyzed, a gas is observed to form at the anode

Answer A (Chlorine)

Question 1 (10 Marks)

10. Electrolysis is

Answer: C

Question 2 (20 marks)

- (a) In a sample of blood at 25°C , $[\text{H}^{+}] = 4.6 \times 10^{-8} \text{ M}$. Determine the molar concentration of OH^{-} and decide if the sample is acidic, basic or neutral.

(5 marks)

$\text{pH} = -\log [\text{H}^{+}]$
 $\text{pH} = -\log (4.6 \times 10^{-8})$
 $\text{pH} = 7.34$
 ≈ 7
- neutral

slightly basic.

$\text{pH} + \text{pOH} = 14$
 $\text{pOH} = 14 - 7.34$
 $= 6.66$
 $\text{pOH} = -\log [\text{OH}^{-}]$
 $6.66 = -\log [\text{OH}^{-}]$
 $[\text{OH}^{-}] = 2.2 \times 10^{-7} \text{ M}$

(4)

Question 2 (20 marks)

- (b) Lactic acid, $\text{C}_3\text{H}_6\text{O}_3$ which is present in sour milk, also gives sauerkraut its tartness is a monoprotic acid. In a 0.1 M solution of its sample, the pH is 2.44 at 25°C . Calculate K_a and $\text{p}K_a$ of lactic acid at this temperature.

(9 marks)

$\text{pH} = 2.44$

$-\log[\text{H}^+] = 2.44$

$[\text{H}^+] = 3.63 \times 10^{-3} \text{ M}$

$x = 3.63 \times 10^{-3} \text{ M}$

$\text{C}_3\text{H}_6\text{O}_3$

initial [M] 0.1

change [M] $-x$

final [M] $0.1 - x$

$\rightleftharpoons \text{H}^+ + \text{C}_3\text{H}_5\text{O}_3^-$

0 0

$+x$ $+x$

x x

$K_a = \frac{[\text{H}^+][\text{C}_3\text{H}_5\text{O}_3^-]}{[\text{C}_3\text{H}_6\text{O}_3]}$

$k_a = \frac{(3.63 \times 10^{-3})^2}{(0.1 - 3.63 \times 10^{-3})}$

$= 1.37 \times 10^{-4}$

$\text{p}K_a = -\log K_a$

$\text{p}K_a = -\log(1.37 \times 10^{-4})$

$\text{p}K_a = 3.86$



Question 2 (20 marks)

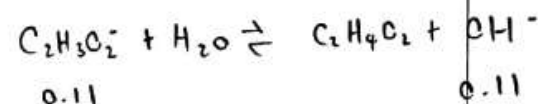
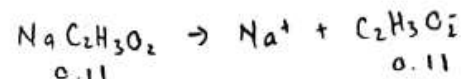
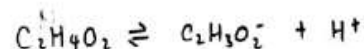
- (c) In order to study the effect of a weakly acidic medium on the rate of corrosion of a metal alloy, a student prepared a buffer solution containing both 0.11 M $\text{NaC}_2\text{H}_3\text{O}_2$ and 0.09 M $\text{C}_2\text{H}_4\text{O}_2$. Calculate the pH of this solution. $K_a = 1.8 \times 10^{-5}$ (6 marks)

$$\text{pH} = \text{p}K_a - \log \left[\frac{\text{Acid}}{\text{base}} \right]$$

$$\text{p}K_a = -\log [K_a]$$

$$\begin{aligned} \text{p}K_a &= -\log [1.8 \times 10^{-5}] \\ &= 4.74 \end{aligned}$$

$$\begin{aligned} \text{pH} &= 4.74 - \log \left[\frac{0.09}{0.11} \right] \\ &= 4.74 - (-0.09) \\ &= 4.83 \end{aligned}$$



6

Question 3 (20 marks)

- (a) Differentiate between galvanic and electrolytic cells. Sketch a simple galvanic cell to support your answer.

(4 marks)

(4 marks)



salt bridge
electrode
solution

Galvanic cell

- $E^{\circ}_{\text{cell}} > 0$
- no electricity supply



electrode

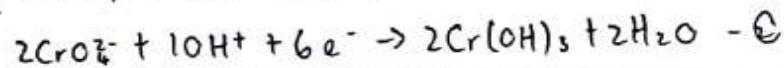
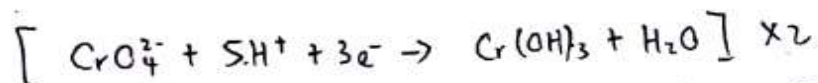
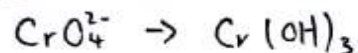
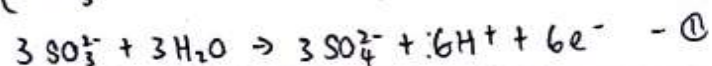
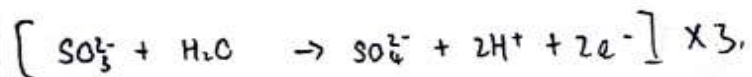
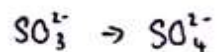
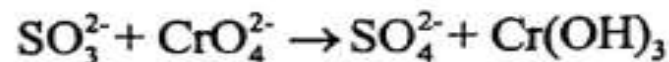
Electrolytic cells

- $E^{\circ}_{\text{cell}} < 0$
- with electricity supply (battery)

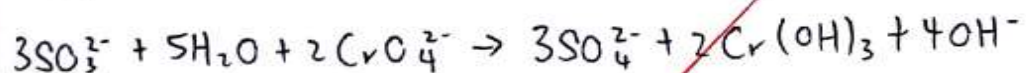
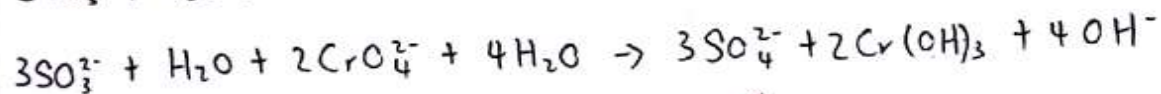
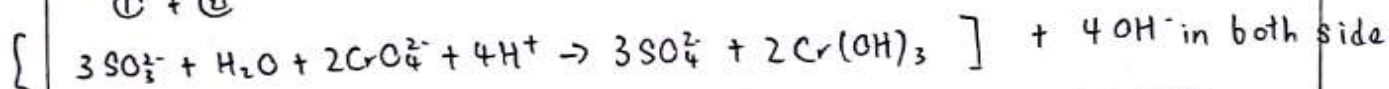
4

Question 3 (20 marks)

(b) Balance the following reaction in basic aqueous solution.

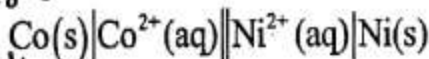
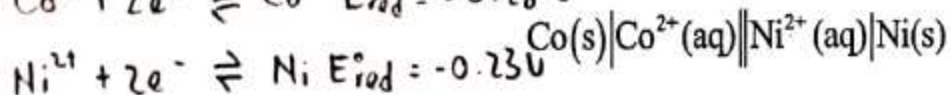
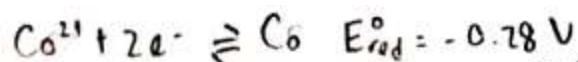


① + ②

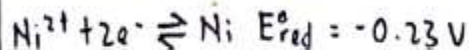
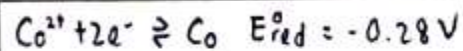


Question 3 (20 marks)

- (c) For the following voltaic cell, write the half reactions, designating their oxidation and reduction processes. Write the cell reaction and calculate the voltage of the cell made from standard electrodes. The cell is

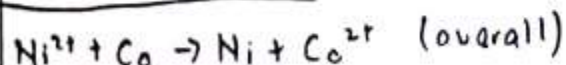
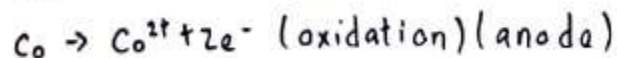
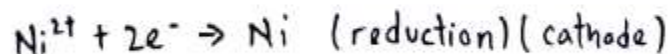


(5 marks)



Cause Ni has higher E_{red}° (more positive).

Ni carry out reduction.



$$E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ}(\text{cathode}) - E_{\text{red}}^{\circ}(\text{anode})$$

$$= -0.23 \text{ V} - (-0.28 \text{ V})$$

$$= 0.05 \text{ V}$$

5

Question 3 (20 marks)

(d) Answer the following questions:

- i. Give two methods of corrosion protections.
- ii. Sketch a diagram showing an electrolytic cell for the electrolysis of aqueous NaCl using inert platinum electrodes.

(4 marks)

i) - coating

- connect a lithium block with that metal by a wire

- painting

ii)

